For all $i \neq k$, $0 \leq i, k \leq n-1$, N_i is an ancestor of N_K if and only if N_i has the smallest priority among all nodes in $N(i, k)$.

We will not prove this lemma here: if you'd like to take the challenge, it can be easily proven by induction.

So, armed with the above lemma, we can compute the probability that the node with the i-th smallest key becomes an ancestor of the node with the k-th smallest key: since we assume that the priorities are drawn from a uniform continuous set, like for instance all real numbers between 0 and 1, then each node in a subset of nodes is equally likely to hold the smallest priority.

Therefore, for each $i \neq k$ we can write the probability of i being an ancestor of k as:

$$P\left(A_k^i\right)_{i \neq k} = \frac{1}{|N\ (i,k)|} = \frac{1}{|k-i|+1}$$

while for $i=k$, instead, the probability is simply 0 (a node can't be its own ancestor).

Replacing these values in the formula for the expected value of D_K, we get:

$$k^* = \frac{m}{n} \cdot ln(2)$$

The middle term of the sum obviously evaluates to 0, while for the first term of the sum, we notice that when $i=0$ the